Statistical Methods & Machine Learning 1

Class #13

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Outline – Experimental Design and Analysis

- Exam 2
- Exam 3 and final class info
- Causal Inference
  - Experiments
  - Counterfactuals
- Randomized Experiment
  - Factorial Designs – One Factor
  - Analysis for Randomized One-Factor Experiments

Exam 2

- Grades should be posted today
- One thing I want to clarify:
  - If a question asks you to interpret the regression coefficients from a model you should interpret the regression coefficients from the model—**not just those where $p < 0.05$**

Exam 3

- Exam 3 will work largely the same as exams 1 and 2.
- It will become available at 12pm next Tuesday (12/3) after our final class.
- Will be **due at 12pm** the following Tuesday.
- Exam 3 will function the same as the prior exams:
  - On your own
  - Take home
  - Open note/open book
  - No AI or LLMs

Group evaluation

- Next week you will also be asked to submit group evals.
  - **<u>These are anonymous.</u>**
- It is a brief form that should be filled out for each of the 2 groups you completed homework assignments in.
- You must complete and submit this form for each group in order to receive full participation credit (5% of your grade).

Class next week

- We will have a little more material to go over—resampling and cross-validation methods.
- Will also have a final lab and quiz.
- Should also have time for a bit of review, so please let me know if there are any topics in particular you would like to discuss.

Causal Inference

Causal Relationship

- Cause: What produces an outcome (IV; Experimental factor [e.g., Treatment vs. Control; dosage)
- Effect: What is produced (DV; Outcome; [e.g., death, high blood pressure])
- **<u>Causation implies covariation AND causal mechanism</u>**
- **<u>Covariation alone is not enough to infer causation</u>**
- Spuriousness: Covariance without causation as causation
  - Waffle house density and divorce rate – confounded by age at marriage
  - Non-causal covariance: # of drownings $\overset{?}{\rightarrow}$ # of Nic Cage movies – covariance is common

Assessing causality

Counterfactual

- What would have happened if all exposed had been unexposed (and all unexposed had been exposed)?
- This is the counterfactual as it's counter to the fact of what actually happened
- Can we actually observe the counterfactual?

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- This is the counterfactual as it's counter to the fact of what actually happened
- Can we actually observe the counterfactual?

No, not without a time machine.

So, what can we do to approximate a counterfactual?

Experiments

Logic of Causal Relationship and Logic of Experiments

- Causal relationship:
  - Cause must precede effect (i.e., temporality)
  - Cause is associated with the effect
  - No other plausible explanation for observed association
- Experiments:
  - Manipulate the exposure or treatment and observe what happens with outcome
  - Minimize possibility of other explanation(s) for effect
  - Variation in the cause is related to variation of effect

Logic of Causal Relationship and Logic of Experiments

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- Experiments:
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  - **Minimize possibility of other explanation(s) for effect**
  - Variation in the cause is related to variation of effect

So, how do we do this without a time machine?

Exchangeability

- To minimize the potential for other explanations of the effect we ensure that the groups are "exchangeable"
- Differences between exposed and unexposed groups are only due to random error
- How do we accomplish this?
  - Design of study/experiment
  - Analytically (to a point)

Exchangeability

- To minimize the potential for other explanations of the effect we ensure that the groups are "exchangeable"
- Differences between exposed and unexposed groups are only due to random error
- How do we accomplish this?
  - **Design of study/experiment** → Randomization is ideal if possible.
  - Analytically (to a point)

Structures in experimental design

- Treatment structure:
  - Set of treatments (or treatment combinations) to compare
  - How to structure treatments:
    - One-way vs. multi-way treatment structure
    - Interactions
    - Fixed vs. Random effects
    - Crossed vs. nested, etc.
- Design structure:
  - Grouping of experimental units into homogenous groups
  - How to assign experimental groups to treatments
    - Balanced vs. non-balanced
    - Randomization
    - Crossover, etc.

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    - Crossover, etc.

Experimental Designs – One factor

Types of one factor designs

Design 1: Basic Treatment vs. Control

R	X	O
R		O

Design 2: Basic Comparing Multiple Treatment Conditions

R	X_A	O
R	X_B	O
R	X_C	O

Design 3: Basic Comparing Multiple Treatments with Control

R	X_A	O
R	X_B	O
R		O

Design 4: Pretest-Posttest Control Group

R	O_1	X	O_2
R	O_1		O_2

Design 5: Pretest-Posttest With Multiple Treatments

R	O_1	X_A	O_2
R	O_1	X_B	O_2
R	O_1		O_2

Types of one factor designs

Randomization

Outcome observed

Treatment assigned

Design 1: Basic Treatment vs. Control

R	X	O
R		O

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Common terminology in experiments

- Factorial design: every possible combination of treatment variables observed in the data
- Balanced design: the number in each treatment group is the same

Will often see these analyses discussed using ANOVA

What is ANOVA?

- ANOVA = analysis of variance
  - AKA regression model with categorical predictor(s)
  - In the lab you will fit models using lm(), aov(), and anova() showing that they are really doing the same thing
- ANOVA models make the same assumptions
  - Errors are Normal IID
  - Zero error mean
  - Constant error variance
  - Errors are normally distributed
- Diagnosis of assumption violations done in the same way
  - Scatterplots of residuals vs. fitted values, QQ-plots of residuals, etc.

One-way ANOVA

- Analyze Y_{ij}, the outcome for a unit j assigned to condition i
  - for $i = 1, \ldots, \alpha$ (i.e., there are α levels of treatment)
  - for $j = 1, \ldots, n_i$ (i.e., there are n_i units assigned to the i^{th} treatment)
  - When $n_i = n$ → "Balanced design"
- Cell means model: $Y_{ij} = \mu_i + \epsilon_{ij}$
  - μ_i: Cell i mean
- Effects model: $Y_{ij} = \mu + \alpha_i + \epsilon_{ij}$
  - μ: Overall mean
  - α_i: Treatment effects

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  - When $n_i = n$ → "Balanced design"
- Cell means model: $Y_{ij} = \mu_i + \epsilon_{ij}$ ← Estimates mean for each treatment level, not overall mean
  - μ_i: Cell i mean
- Effects model: $Y_{ij} = \mu + \alpha_i + \epsilon_{ij}$ ← Estimates overall mean and the deviation each treatment from overall mean
  - μ: Overall mean
  - α_i: Treatment effects

One-way ANOVA

- One-way ANOVA tests the hypothesis:
  - $H_0: \alpha_1 = \alpha_2 = \cdots = \alpha_a \; vs. H_A: At\ least\ one\ of\ the\ a\ is\ different$, or
  - $H_0: \mu_1 = \mu_2 = \cdots = \mu_a \; vs. H_A: At\ least\ one\ of\ the\ \mu\ is\ different$

Pairwise Difference Tests

Pairwise difference tests with two-level treatment

- One factor (i.e., categorical predictor) with two levels
  - E.g., LDL cholesterol for no wine (*T*) vs. wine (*C*)
  - Hypothesis: ?

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- How can we test this hypothesis?

Pairwise difference tests with two-level treatment

- One factor (i.e., categorical predictor) with two levels
  - E.g., LDL cholesterol for no wine (*T*) vs. wine (*C*)
  - Hypothesis: $H_0: \mu_T = \mu_C \; vs. H_A: \mu_T \neq \mu_C$
- How can we test this hypothesis?
  - Series of tests covered in Class 2!
    - Known vs. unknown variance (Z vs. t tests)
    - Equal vs. unequal variance
    - Independent vs. dependent sample

Pairwise difference tests with >2 levels of treatment

- One factor with more than two levels
- E.g., LCL level for no wine (*A*), 2oz wine (*B*), and 4oz wine (*C*)
- Hypotheses: ?

Pairwise difference tests with >2 levels of treatment

- One factor with more than two levels
- E.g., LCL level for no wine (*A*), 2oz wine (*B*), and 4oz wine (*C*)
- Hypotheses:
  - $H_0: \mu_A = \mu_B \ vs. H_A: \mu_A \neq \mu_B$
  - $H_0: \mu_A = \mu_C \ vs. H_A: \mu_A \neq \mu_C$
  - $H_0: \mu_B = \mu_C \ vs. H_A: \mu_B \neq \mu_C$

Pairwise difference tests with >2 levels of treatment

- One factor with more than two levels
- E.g., LCL level for no wine (*A*), 2oz wine (*B*), and 4oz wine (*C*)
- Hypotheses:
  - $H_0: \mu_A = \mu_B \ vs. H_A: \mu_A \neq \mu_B$
  - $H_0: \mu_A = \mu_C \ vs. H_A: \mu_A \neq \mu_C$
  - $H_0: \mu_B = \mu_C \ vs. H_A: \mu_B \neq \mu_C$

May also see written as:
$H_0: \mu_A - \mu_B = 0 \ vs. H_A: \mu_A - \mu_B \neq 0$

Pairwise difference tests with >2 levels of treatment

- One factor with more than two levels
- E.g., LCL level for no wine (A), 2
- Hypotheses:
  - $H_0: \mu_A = \mu_B \ vs. H_A: \mu_A \neq \mu_B$
  - $H_0: \mu_A = \mu_C \ vs. H_A: \mu_A \neq \mu_C$
  - $H_0: \mu_B = \mu_C \ vs. H_A: \mu_B \neq \mu_C$

After seeing the hypotheses tested when the treatment has >2 levels, is there anything we need to consider?

Pairwise difference tests with >2 levels of treatment

- Multiple hypotheses!
  - Likelihood of erroneously rejecting correct null hypothesis (Type 1 Error) increases
  - Have to account for this via correction:
    - Bonferroni correction
    - Tukey's honest significant difference test
    - False discovery rate

Randomized Multi-way Design Structure

Design Options with Multiple Factors - 1

Two treatment factors: *A* and *B*

Fully Crossed - $A \times B$

R	X_{A1B1}	O
R	X_{A1B2}	O
R	X_{A1B3}	O
R	X_{A2B1}	O
R	X_{A2B2}	O
R	X_{A2B3}	O

Design Options with Multiple Factors - 2

Three treatment factors: *A*, *B*, and *C*

Fully Crossed - $A \times B \times C$

R	X_{A1B1C1}	O
R	X_{A1B1C2}	O
R	X_{A1B2C1}	O
R	X_{A1B2C2}	O
R	X_{A2B1C1}	O
R	X_{A2B1C2}	O
R	X_{A2B2C1}	O
R	X_{A2B2C2}	O

Two-way Treatment Structure Models

Two-way ANOVA

Analysis of Y_{ijk}, where

$i = 1, \cdots, I$: levels in factor A

$j = 1, \cdots, J$: levels in factor B

$k = 1, \cdots, n_{ij}$: subjects assigned to the level ij of combined factors A and B

$\sum_{i=1}^{I} \sum_{j=1}^{J} n_{ij} = n_{\bullet\bullet}$: total sample size

$n_{ij} = n \rightarrow$ Balanced design

$n_{ij} = 1 \rightarrow$ No replication

Y_{ijk} can be expressed with a cell means model and an effects model

Two-way ANOVA: Cell Means Model

Cell means model:

$Y_{ijk} = \mu_{ij} + \epsilon_{ikk}$

μ_{ij}: Cell mean for *ij*th level of the combined factors *A* and *B*

OLS estimator for μ_{ij}:

$$\hat{\mu}_{ij} = \hat{\mu} + \hat{\alpha}_i + \hat{\beta}_j + \hat{\gamma}_{ij} = \frac{\sum_{k=1}^{n_{ij}} Y_{ijk}}{n_{ij}}$$

Assumes: $\epsilon_{ijk} \sim N(0, \sigma^2)$

Two-way ANOVA: Effects Model

Effects model:

- $Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij}\epsilon_{ijk}$
  - μ: Overall mean
  - α_i: the main effect for i^{th} level of factor A
    - $\sum_{i=1}^{I} \alpha_i = 0$
  - β_j: the main effect for j^{th} level of factor B
    - $\sum_{j=1}^{J} \beta_j = 0$
  - γ_{ij}: the interaction effect between the i^{th} level of factor A and the j^{th} level of factor B
    - $\sum_{i=1}^{I} \gamma_{ij} = \sum_{j=1}^{J} \gamma_{ij} = 0$

Two-way ANOVA: Effects Model

Effects model:

- $Y_{ijk} = \mu + \alpha_i$
  - μ: Overall me
  - α_i: the main
    - $\sum_{i=1}^{I} \alpha_i = 0$
  - β_j: the main effect for j^{th} level of factor B
    - $\sum_{j=1}^{J} \beta_j = 0$
  - γ_{ij}: the interaction effect between the i^{th} level of factor A and the j^{th} level of factor B
    - $\sum_{i=1}^{I} \gamma_{ij} = \sum_{j=1}^{J} \gamma_{ij} = 0$

- Assumes $\epsilon_{ijk} \sim N(0, \sigma^2)$
- If no interaction is expected, you can use:
  - $Y_{ijk} = \mu + \alpha_i + \beta_j + \epsilon_{ijk}$

Fixed vs. random effects

Experimental Factor: Fixed vs. Random

- Fixed effect
  - α fixed treatments specifically chosen by the experimenter
  - Conclusions apply only to the factor levels considered in the analysis
- Random effect
  - α treatments randomly chosen from a larger populations of treatments
  - Conclusions extend to similar treatments not explicitly considered in the analysis

Fixed vs. Random Effects in One-Way Model

- Y_{ij} with $i = 1, \dots, \alpha$ treatment levels and $\mathrm{j} = 1, \dots, n_i$ subjects
- When the treatment levels in factor A include the entire population of all possible levels, or are selected purposely, A is a **fixed factor** (AKA fixed effect)
- When the treatment levels in factor A include a random sample from a population of all levels, A is a random factor (AKA random effect)
- These are different model specifications depending on the nature of A

Fixed vs. Random Effects Underlying Models

Y_{ij} under **fixed effects**:	Y_{ij} under **random effects**:
$E(Y_{ij}) = \mu + \alpha_i$	$E(Y_{ij}) = \mu \quad \because \alpha_i \sim N(0, \sigma_\alpha^2)$
$V(Y_{ij}) = \sigma^2 = \sigma_\epsilon^2$	$V(Y_{ij}) = \sigma^2 = \sigma_\alpha^2 + \sigma_\epsilon^2$
$C(Y_{ij}, Y_{ij'}) = 0 \rightarrow$ "Crossed"	$C(Y_{ij}, Y_{ij'}) = \sigma_\alpha^2 \rightarrow$ "Nested"
$C(Y_{ij}, Y_{i'j'}) = 0$	$C(Y_{ij}, Y_{i'j'}) = 0$
$Cor(Y_{ij}, Y_{ij'}) = 0$	$Cor(Y_{ij}, Y_{ij'}) = \dfrac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma_\epsilon^2}$
Interested in estimating α_i and/or differences in α_i	Interested in estimating variance components (i.e., contributions of σ_α^2 and σ_ϵ^2 in relation to σ^2)

Fixed vs. Random Effects Underlying Models

Y_{ij} under **fixed effects**:	Y_{ij} under **random effects**:
$E(Y_{ij}) = \mu + \alpha_i$	$E(Y_{ij}) = \mu \quad \because \alpha_i \sim N(0, \sigma^2_\alpha)$
$V(Y_{ij}) = \sigma^2 = \sigma^2_\epsilon$	$V(Y_{ij}) = \sigma^2 = \sigma^2_\alpha + \sigma^2_\epsilon$
$C(Y_{ij}, Y_{ij'}) = 0 \rightarrow$ "Crossed"	$C(Y_{ij}, Y_{ij'}) = \sigma^2_\alpha \rightarrow$ "Nested"
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Interested in estimating α_i and/or differences in α_i	Interested in estimating variance components (i.e., contributions of σ^2_α and σ^2_ϵ in relation to σ^2)

This has important implications for the hypothesis being tested

Fixed vs. Random Effects ANOVA -1

ANOVA table with one-way treatment structure with balanced design:

Fixed Effect

Source	*df*	$E(MS)$	MS_{fixed}	F
Treatments A	$a-1$	$E(MS_A)$	$\sigma_\epsilon^2 + \frac{1}{a-1}\sum_i n_i(\alpha_i - \bar{\alpha})$	$\frac{MS_A}{MS_E}$
Error	$na-a$	$E(MS_E)$	σ_ϵ^2	
Total	$na-1$			

Random Effect

Source	*df*	$E(MS)$	MS_{random}	F
Treatments A	$a-1$	$E(MS_A)$	$\sigma_\epsilon^2 + \frac{1}{a-1}(\sum_i n_i - \frac{\sum_i n_i^2}{\sum_i n_i})\sigma_\alpha^2$	$\frac{MS_A}{MS_E}$
Error	$na-a$	$E(MS_E)$	σ_ϵ^2	
Total	$na-1$			

Fixed vs. Random Effects ANOVA -2

ANOVA table with one-way treatment structure with balanced design:

Fixed Effect

Hypothesis tested with $F = \dfrac{MS_A}{MS_E}$

$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_a$ vs. H_A: Not H_0

Random Effect

Hypothesis tested with $F = \dfrac{MS_A}{MS_E}$

$H_0 : \sigma^2_\alpha = 0$ (i.e., does σ^2_α play a role in σ^2?) vs. H_A: $\sigma^2_\alpha > 0$